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25 March 1983
SUPERSEDING
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21 March 1963

MILITARY SPECIFICATION

MOTOR-GENERATOR PU 335( )/MPM-25

This specification is approved for use by the Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of motor-generator designated Motor-Generator PU 335( )/MPM-25 (See 6.1 and 6.3).

2. APPLICABLE DOCUMENTS

2.1 Government Documents

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specification and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, DELET-R-S, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of.

MIL-W-6858 - Welding: Aluminum, Magnesium, Nonhardening Steels or Alloys, and Titanium, Spot, Seam, and Stitch.

MIL-M-10578 - Metal Conditioner and Rust Remover (Phosphoric Acid Base)

MIL-M-13231 - Marking of Electronic Items.

MIL-F-14072 - Finishes for Ground Signal Equipment.

MIL-E-55585 - Electronics Equipment and Parts Packaging of

STANDARDS

MILITARY

MIL-STD-22 - Welded Joint Designs.

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-169 - Extreme - Temperature Cycle.

MIL-STD-252 - Wired Equipment, Classification of Visual and Mechanical Defects.

MIL-STD-461 - Electromagnetic Emissions and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

MIL-STD-726 - Packaging Requirements Code

MIL-STD-810 - Environmental Test Methods

GOVERNMENT DRAWINGS

SC-A-46439 - List of Accessories for Package Tester.

SC-DL-48271 - Motor-Generator Set PU-335( )/MPM-25

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

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3. REQUIREMENTS

3.1 Construction. The motor-generator shall be constructed and assembled in accordance with Drawing - SC-DL-48271.

3.2 First Article. When specified, two samples shall be subject to first article inspection (see 4.3 and 6.2).

3.3 Processes.

3.3.1 Soldering.

3.3.1.1 Acid or acid salts. No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.3.1.2 Process. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.3.2 Welding. Wherever practicable, welded joints shall conform to MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Spot welds shall conform to MIL-W-6858.

3.3.2.1 Cleaning prior to welding. Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.3.2.2 Process. Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum.

3.3.2.3 Cleaning after welding. Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

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3.3.3 Wiring and cabling. Wiring and cabling shall be neat and sturdy.

3.3.3.1 Slack. Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

a. To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

b. To enable parts to be removed and replaced during servicing without disconnecting other parts.

c. To provide for at least two replacements of the part to which the wire or cable is connected.

d. To facilitate field repair of broken or cut wires.

e. To prevent chafing or breaking of individual wires due to repeated flexing of hinged parts.

3.3.3.2 Protection. Wires shall not be bent sharply where they enter insulation material.

3.3.3.3 Support. Wire and cable shall be properly supported and secured to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable during and after subjection of the equipment to specified service conditions or after service or repair of the equipment in a normal manner. Twine or tape shall not be used for securing wire and cable.

3.3.3.4 Clearance. Clearance between wires or cables shall be provided to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.3.3.5 Splicing. Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.3.4 Tropicalization of material. Material shall be treated as follows:

3.3.4.1 Treating materials. Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.3.4.2 Toxicity. Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

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3.3.4.3 Flexibility. Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.3.4.4 Statement of treatment. The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2.)

3.3.5 Cleaning.

3.3.5.1 Parts. After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.3.5.2 Units. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.3.6 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 4.4.)

3.3.7 Marking.

3.3.7.1 General. Marking shall conform to MIL-M-13231. (See 4.4.)

3.3.7.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.3.7.3 Serial Numbers. Motor-Generator PU-335( )/MPM-25 shall be serial numbered.

3.4 Service conditions. The equipment shall meet the following service conditions-- where a test is referenced, meeting the test shall be considered as compliance with the requirement:

3.4.1 Operation. Continuous use for a period of five years, under the worldwide environmental conditions specified in 3.4.2 through 3.4.5 with a duty cycle of 16 hours on and 8 hours off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement. Approval of the first article sample (see 3.2) shall be considered as compliance with this requirement.

3.4.2 Temperature (see 4.6).

a. Operating: Ambient temperature in the range of +65°C to -40°C (+150°F to -40°F). (The 65°C temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed four hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

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b. Nonoperating: Exposure in the range of +71°C to -62°C (+160°F to -80°F.); exposure of the high temperature extreme not to exceed four hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.4.3 Relative humidity (see 4.7). Daily exposure up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for four hours.

3.4.4 Bounce. The equipment shall meet full specification requirements as specified herein and shall show no physical damage except minor surface abrasions when subjected to the bounce test of 4.8.

3.4.5 Vibration. The amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second. (See 4.23.)

3.4.6 Shock, bench-handling. The equipment shall meet full specification requirements as specified herein and shall show no physical damage after subjection to shocks simulating those normally occuring during servicing.

3.5 Performance

3.5.1 Power source. The equipment shall be reconnectable to operate from a 60 cycle per second (cps) ±5 percent single phase, 115/230 volts ±10 percent alternating current (ac) two wire source. (See 4.10.)

3.5.2 Power output. The generator shall be capable of delivering 500 watts ±10 percent 3 phase at 120 ±10 percent volts between phase to neutral and at 208 ±10 percent volts phase to phase (see 4.10).

3.5.3 Frequency. The frequency at the voltages specified in 3.4.2 shall be 400 cps ±5 percent. (see 4.11).

3.5.4 Phase displacement. The displacement between phases shall be 120 degrees ±5 percent. (See 4.12.)

3.5.5 Waveform. The output waveform of the motor-generator shall be a sine-wave with a maximum of 5 percent harmonic distortion on each phase. (See 4.13.)

3.5.6 Dynamic balance. The free vibration of the motor-generator under any load between zero output current and rated output current shall not exceed 0.002 inch total excursion. (See 4.14.)

3.5.7 Temperature rise. The temperature rise of the motor, under full load, shall not exceed 55°C. (132°F) The temperature rise of the generator shall not exceed 40°C (104°F). (See 4.15.)

3.5.8 Insulation resistance. The insulation resistance shall be 100 megohms, minimum. (See 4.16.)

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3.5.9 Dielectric strength. The windings shall withstand 1460 volts, direct current (dc), minimum, without breakdown. (See 4.17.)

3.5.10 Life intermittent duty. The motor-generator shall meet full specification requirements as specified in 3.5.1 through 3.5.9 after operation with nominal input for periods of one hour on at full load, and 1/2 hour off for a total of 500 hours. (See 4.18.)

3.5.11 Final operation. Just prior to packaging, the equipment shall be examined visually and electrically as specified in 4.19.

3.6 Electromagnetic Interference. The equipment shall comply with class IIIa as specified in table A-11-MIL-I-11748 of Notice 4 to MIL-STD-461A. (See 4.20)

3.7 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.21.) Individual items shall not be hand-picked for fit or performance; however, matched pairs may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.8 Technical literature, tools, and running spare parts. Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2.)

3.9 Workmanship. The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.3.1 Soldering	3.3.3 Wire and cable
3.3.2 Welding	3.3.5 Cleaning

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The manufacturer shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

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a. First article inspection (See 4.3.)

b. Quality conformance inspection (See 4.4 and 4.5)

c. Packaging Inspection (See 4.24)

4.3 First article inspection. First article inspection shall be performed at a laboratory acceptable to the Government (See 6.2) on sample units produced with equipment and procedures normally used in production. First article inspection shall consist of tests specified in table I, and the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for groups A, B, and C (See tables II,III and IV respectively).

Table I. First article inspection

Inspection (For additional first article inspection see 4.3.)	Req para	Insp para
Temperature	3.4.2	4.6
Moisture-resistance	3.4.3	4.7
Electromagnetic Interference	3.6	4.20

4.4 Acceptance inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before packaging:

Item	Where required
Finish	3.3.6
Marking	3.3.7

4.5 Acceptance inspection of equipment before packaging. The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.7.) Test equipment for Government verification inspection shall be made available by the contractor.

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4.5.1 Group A inspection. This inspection, including sampling, shall conform to table II and the ordinary inspection procedures of MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government except that the final operation inspection (See 4.19) shall be last.

Table II. - Group A inspection

Inspection	Req para	Insp para	AQL Major	AQL Minor
Visual and mechanical--	3.8.8	4.22	2.5 dphu 1/	10 dphu
Electrical				
Power output--------	3.5.2	4.10	-- 2.5 dphu for the group	2/
Frequency----------	3.5.3	4.11		
Phase displacement--	3.5.4	4.12		
Insulation resistance-	3.5.8	4.16		
Dielectric strength---	3.5.9	4.17		
Final operation--------	3.5.11	4.19	1%	

1/ Defects per hundred units.

2/ All electrical defects are considered major.

4.5.2 Group B inspection. This inspection including sampling, shall conform to table III and to the special procedures for small-sample inspection of MIL-STD-105. The AQL shall be 6.5 percent defective and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met group A inspection.

Table III. - Group B inspection

Inspection	Req Para	Insp Para
Dynamic balance----------	3.5.6	4.14
Interchangeability----------	3.7	4.21

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4.5.3 Group C inspection. This inspection shall be listed in table IV, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

Table IV. - Group C inspection

Inspection	Req para	Insp para
Bounce	3.4.4	4.8
Shock, bench-handling	3.4.6	4.9
Waveform	3.5.5	4.13
Temperature rise	3.5.7	4.15
Life	3.5.10	4.18
Vibration	3.4.5	4.23

4.5.3.1 Sampling for inspection of equipment. Two units out of each 200 produced or fraction thereof shall be selected for group C inspection, without regard to their quality, except that the units inspected at the start of the contract shall be selected from the first units produced.

4.5.3.2 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and to the qualifying activity.

4.5.4 Reinspection of conforming group B and group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are subjected to and pass group A inspection after repair of all visible damage.

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4.6 Temperature test (see 3.4.2). The equipment shall be subjected to the temperature cycle shown on MIL-STD-169. The equipment shall be tested at steps 1, 3, 5, 8 and 10 for the following performance:

Paragraph	Measurement	Performance
4.10	Power output	As specified in 3.5.2
4.11	Frequency	As specified in 3.5.3
4.16	Insulation resistance	As specified in 3.5.8
4.17	Dielectric strength	As specified in 3.5.9

The test for waveform (see 4.13) shall also be performed. This test shall be made prior to placing the unit into the chamber and then repeated after removal from the chamber. The results shall be compared to determine that the performance meets full specification requirements.

4.7 Moisture-resistance test (see 3.4.3). The equipment shall be tested as follows:

4.7.1 Test conditions.

a. Do not move equipment from the humidity chamber for measurements.

b. Start measurements not more than five minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.7.2 Test method.

a. Dry at 54°C ±3°C (130°F ±5°F) for 24 hours

b. Condition at 25°C ±2°C (77°F ±5°F) and 40 to 50 percent relative humidity for 24 hours.

c. Take measurements as specified in 4.7.3

d. Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-810. The measurements specified in 4.7.3 shall be made during the periods shown on the standard and shall comply with 4.7.3.

e. After cycling has been completed, condition the equipment for 24 hours at 25°C ±5°F) and 40 to 60 percent relative humidity. No repair or replacement of parts shall be made. The equipment shall meet full specified performance for those measurements specified in 4.7.3.

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4.7.3 Performance. Measurements made during test and specified performance during cycling shall be as follows:

Paragraph	Measurement	Performance during cycling
4.10	Power output	As specified in 3.5.2
4.11	Frequency	As specified in 3.5.3
4.16	Insulation resistance	As specified in 3.5.8
4.17	Dielectric strength	As specified in 3.5.9

The test for waveform (see 4.13) shall also be performed. This test shall be made prior to placing the unit into the chamber and then repeated after removal from the chamber. The results shall be compared to determine that the performance meets full specification requirements.

4.7.4 Failure. If any equipment fails to meet the performance specified in 4.7.3 during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning as specified in 4.7.2(e), it does not pass the test.

4.8 Bounce test; class F units, (see 3.4.4). The equipment shall be tested on the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

a. Cover the tester bed with a panel of 1/2 inch plywood with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at six-inch intervals around all four edges and at three-inch intervals in a six-inch square in the center.

b. Place the equipment prepared for field transport, on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than three inches and not less than one inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the backstop is not restrained.

c. Operate the package tester, shafts in phase, for a total of three hours at 284 $\pm$2 rpm. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

4.9 Shock test; bench-handling (see 3.4.6.). The equipment shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, two-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

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4.10 Power Output. With the equipment connected to a power source as specified in paragraph 3.5.1 the output shall be measured to determine compliance with 3.5.2

4.11 Frequency. A frequency indicator with an accuracy of 0.5 percent shall be connected to the generator output. The frequency measurements shall conform to 3.5.3.

4.12 Phase displacement. The output of the generator shall be connected to an oscilloscope so as to display the voltage variation with time in such a manner that the phase displacement in degrees can be measured for conformance to 3.5.4.

4.13 Waveform. The waveform of the voltage developed on each of three phases of the generator output shall be recorded on an oscillograph. A detailed examination of the entire trace shall be made and instantaneous voltages measured at no less than 15 degree intervals. The results shall conform to 3.5.5.

4.14 Dynamic balance. The completely assembled equipment shall be elastically mounted to allow vibration in all axes. The elastic members of the mounting may be either compression or tention. The weight of the equipment, when not rotating, shall cause the mounting to be compressed (stretched) to at least the amount shown below:

Rated speed, rpm	Minimum displacement, inch
900	1
1,200	9/16
1,500	3/8
1,800	1/4
3,600	1/16
7,200	1/64

For any rated speed other than shown above, the minimum compression (tension), which is inversely proportional to the square of the motor speed, shall be interpolated. The static compression (tension) shall be not greater than 50 percent of the original length of the compression (tension) member. With the equipment mounted and rotating, the load shall be applied and varied continuously between one-tenth and full value of the rated load current (see 3.5.2). The equipment shall also be operated at no-load conditions. Vibration at the bearing housing shall be measured with a vibration indicator in the direction giving the maximum amplitude. The mounting shall be isolated to the extent necessary to allow vibration amplitudes of the bearing housing, with the equipment mounted but not rotating, not greater than one-tenth of the maximum allowable amplitude for the rotating equipment. The results shall conform to 3.5.6.

4.15 Temperature rise, (see 3.5.7).

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4.15.1 Motor. The temperature rise of the armature winding of the motor shall be determined by resistance measurements in an area free of drafts and direct thermal radiation, while the equipment is mounted to material of low thermal conductivity such as wood or porcelain. Two commutator bars (on the hottest winding, if possible) shall be marked for identification. The resistance (r) between these marked bars shall be determined with the brushes removed. The equipment shall be cycled until there is no change in the maximum temperature reached during a cycle for three successive duty cycles (see 3.5.10). Resistance measurements shall be made immediately at the end of the "on" portion of the duty cycle. The temperature of the frame of the motor, measured on the external surface of the yoke at the center of the pole shoes, shall be determined by thermocouple or thermometer. The resistance between the marked commutator bars shall be measured at intervals of not more than 30 seconds immediately after the equipment has been stopped, and shall be extrapolated back to the resistance at time of shutdown for determination of R. The temperature rise of the armature winding shall be computed using the following formulas:

$$\Delta T = \frac{R-r}{r} (t + 234.5) - (T - t)$$

Where: ΔT = Armature temperature rise in degrees centigrade.

R = Hot resistance in ohms at time of shutdown.

r = Cold resistance in ohms.

t = Temperature of equipment at the time of measurement of 4 in degrees of centigrade.

T = Mean ambient temperature during final quarter of test in degrees centigrade.

The ambient temperature shall vary by not more than $2^{\circ}C$ ($3^{\circ}F$) during the final quarter of this test. The results shall conform to 3.5.7.

4.15.2 Generator. Repeat the test specified in 4.15.1 except that the generator shall be tested. The results shall conform to 3.5.7.

4.16 Insulation resistance. All brushes may be removed from each commutator and the equipment may be disassembled sufficiently to unground each winding in turn. Measurements shall be made between the frame and each winding which is not permanently connected to the frame by soldering or welding, with all other windings connected to the frame. The insulation resistance shall be measured for conformance to 3.5.8.

4.17 Dielectric strength, (see 3.5.9). The capacitors connected to the circuit under test shall be disconnected so that they will not be stressed during test. All brushes may be removed from each commutator and the equipment may be disassembled sufficiently to unground each winding in turn. The test voltage shall be 1460 vdc, and shall be applied for not less than five seconds nor more than

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one minute. The test voltage shall be applied between the frame and each winding which is not permanently connected to the frame by soldering or welding, with all other windings connected to the frame. The insulation resistance shall be measured as specified in 4.16. The results shall conform to 3.5.9.

4.18 Life test. (see 3.5.10). The motor-generator shall be operated with nominal input for a period of one hour on, with full load, and 1/2 hour off for a total of 500 hours (total of 333 cycles). During the 500 hour period, the output, frequency, temperature rise, insulation resistance, and dielectric strength shall be checked periodically, (see 4.10, 4.11, 4.15, 4.16 and 4.17 respectively).

4.19 Final operational test. Prior to packaging, the equipment shall be checked visually for damage, loose parts and completeness. All electrical circuits shall be checked for apparent malfunction. (See 3.5.11).

4.20 Electromagnetic Interference. The equipment shall be tested for compliance with para 3.6 using the test methods and procedures of Notice 3, MIL-STD-462.

4.21 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.7. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of the case, and internal dimensions of the cover.

b. Location of hinges, clamp, and fasteners on case and cover.

4.22 Visual and mechanical inspection. Equipment shall be examined for the defects listed in MIL-STD-252 and table V.

Table V. - Classification of visual and mechanical defects

Classification	Defect
Major	Clamp, hinge or lid fastener attached to wrong side of case. Cover missing or inoperative. Control inoperative.
Minor	Cover improperly assembled. Difficulty in getting good contact due to play or backlash. Does not operate smoothly and freely.

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4.23 Test for internal vibration. Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.4.5.

a. Secure the equipment directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate the observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

b. Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excussion of the applied vibration shall be not less than 0.030 inch.

c. In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

d. Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.24 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging

b. Quality Conformance Inspection of Packaging

4.24.1 First Article Inspection of Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.24.2 Quality Conformance Inspection of Packaging.

4.24.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.24.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.24.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table VI, "PACKING INSPECTION PROVISIONS." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality Level of 4.0 percent defective.

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TABLE VI

PACKING INSPECTION PROVISIONS

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted.	Visual
109	Marking omitted, incorrect, or illegible	Visual

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5. PACKAGING

5.1 Preservation. Preservation requirements shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 1Ø-1-1-ØØ-00-NS-X-ED-Ø-00-A

5.1.2 Level B. 1Ø-1-1-ØØ-00-NS-X-ED-Ø-00-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES.

6.1 Intended use. This unit is intended for use as a power supply for the Radar Computer CP-212( )/MPQ4, when the computer is being bench tested.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level A or level B preservation and packing (see section 5)

c. First article inspection:

First article pack(s) as follows:

Makeup of pack(s)
Number of each kind of pack to be submitted
Inspection to be performed thereon.

d. Marking and shipping of samples

e. Place of final inspection

f. Technical literature required (See 3.8)

g. Quantity of tools and running spare parts required (See 3.8)

h. Submission of the statement of treatment referenced in 3.2.4.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

i. When first article inspection, rough handling tests are not required.

j. When first article packaging inspection test reports require Acquisition Activity approval prior to production unit packing.

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6.3 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: PU-335W/MPM-25. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2.)

6.5 Level B preservation. When a level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.6. Environmental. Environmental polution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.7 Verification inspection. Verification by the Government (see 4.5) will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definite quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

a. Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by MIL-STD-105.

b. Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of MIL-STD-105.

c. Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

Custodian:

Army - ER

Preparing Activity:

Army - ER

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-55227A (ER)	Military Specification, Motor Generator PU 335 ()/ MPM-25

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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